

## **Environmental dangers of AI**

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### **Abstract**

Artificial intelligence has been a game changer in this decade and has changed a lot of business processes and even everyday lives, where people use it for source of specific information, while some like to play with AI to generate fun images. One of the biggest concerns about the use of AI tools is regarding „the machine replacing humans “or “machine stealing someone’s art”, while only a few realise the true harms AI is causing. Most people take generative artificial intelligence for granted and think it does not come without consequences.

This article aims to describe the environmental consequences, more specifically higher energy and water consumption, carbon emissions, all related with usage of AI tools and reliance of AI technologies on rare minerals. By extracting figures and data from relevant articles and reports, this research will present the rise in negative ecological trends associated with the usage of AI and how can developers and engineers

Not only will this research dig deeper into the harms, but it will as well address potential solutions in harm reduction, and what steps should be taken by taking collective responsibility to transition to greener practices.

Keywords: AI, ecology, technology, harm reduction, environment.

## **1 Introduction**

Generative AI gives the machines an ability to create content, ranging from text and images to audio and video. This marks a significant change from traditional AI, which primarily focused on analyzing or classifying existing data. Since the DALL-E and GPT models were made public, the usage of generative artificial intelligence is on the rise for various of reasons, either it being for business, education or even fun. This sparked many ethical discussions and raised many concerns about its usage. Most of the concerns associated with generative AI are regarding AI bots replacing humans in some business processes and causing a shock on the labour market, or plagiarism, where the creations of artists are stolen by the AI and with slight modifications are presented to the user as its own. However, one issue is massively overlooked by the general public that has long lasting consequences for everyone: the harm to the environment.

Among the reasons why general public takes AI for granted is because they have one big misconception: AI being digital; thus, not being able to be unsustainable for the environment. For an AI to exist, it requires a machine that uses specific rare materials, an energy source to power the machine, and resources to cooldown the machine. The servers hosting and enabling AI operations need to physically exist in order to produce output. With increasing popularity of ChatGPT, new players have entered the AI market shortly after; Google Gemini, Microsoft Copilot and DeepSeek, along with companies developing their internal generative tools, who will all strive to outperform the competitors and provide the best service, which might come at the further cost of environment.

The main question that I will be answering is: “How exactly does AI cause harm to the environment?”. With this research, I will dig deeper into the problem of AI and how exactly do GPT models harm the environment. I will be reviewing literature and analyse secondary qualitative data to present the harmful practices associated with the development and usage of AI. I will break down the research into three main categories and areas: material extraction, energy usage and carbon emissions and water usage, which will provide analytical clarity to discover patterns, trends, causes and effects.

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## **2 Brief background behind Generative AI**

### **2.1 First steps towards Generative AI**

Generative AI is not just a recently thought out concept that was immediately invented. The development of generative AI is the result of decades of research and development, and has been achieved by advancements in computing power, data availability, and innovative algorithmic design.

Generative AI can be traced back to the mid-20th century, with early explorations into machine learning and neural networks. Alan Turing's work on machine intelligence and the Turing Test (originally named Imitation Game by Turing) in the 1950s laid some of the conceptual groundwork. In the 1960s, Joseph Weizenbaum's ELIZA, an early chatbot, demonstrated the potential for machines to imitate human conversation, through relatively simple pattern matching techniques. These innovations highlighted the possibility of machines generating outputs that resemble human created content. During the 1970s, agencies and government stopped funding research due to no development, but private businesses decided to take the matter into their own hands. Businesses benefited from computers and valued the efficiency they brought them with machine learning and they have decided to fund researches instead. The development of generative AI has grown steadily, and in late 2000s, online chatbots and virtual assistants made its debuts. But in 2014, the concept of generative adversarial network (GAN) was shown, which uses two neural networks where one would work as a generator and the other as a discriminator, which would distinguish between real and generated data. This discovery made AI feel more human-like when generating content and marked a significant step towards Generative Pre-Trained Transformer (GPT), which serves as a base for ChatGPT. (Foote, 2024)

## 2.2 Development of GPT

One of the most significant developments in the field of AI is the Generative Pre-trained Transformer (GPT) model developed by OpenAI and it has formed the AI we know of it today. These models are based on deep learning, a type of machine learning, using a type of neural network architecture, which uses billions of parameters and connections, called the Transformer, which was introduced in a 2017 paper titled “Attention Is All You Need.”. The first model was released in 2018, and it has consisted of 117 million parameters, and it showed immense power AI could hold. Only a year after, OpenAI has released a significantly upgraded GPT-2 with 1.5 billion parameters, and its public release was delayed due to debates within OpenAI whether the product will be misused. But OpenAI did not stop there and has developed even bigger and more groundbreaking GPT-3 model, which could generate much more than simple texts, like emails, articles, and poetry. On top of the much more creative language, it could generate programming codes and even images based on the textual description (Marr, 2023). The latest model in the GPT series is the GPT-4 which has several versions released, and the popular ChatGPT is running on GPT-4o version.

## 2.3 Data Centres

Data centres are facilities holding IT infrastructure, which not only enable digital processes, but carry out their operations as well. The infrastructure consists of computers, servers, data storage systems, network equipment which connects them all and cooling mechanisms that prevents the hardware from malfunctioning. (Gantz, 2014).

AI has a strong connection with data centres, and it could be considered the birthplace and where the magic takes place. Data centres are crucial for training and developing AI models, as data centres provide the space and hardware with large computational power strong enough to process large quantities of data that can be as large as several petabytes worth, along with strong network that allows quick data exchange between the computers and servers (Hunter, 2025). As mentioned above, neural networks, programs which make the generative AI possible have billions, some even trillions of parameters and connections, require a host server strong enough to support their operations and train the AI for generating content. Not only do data centres work as a place where AI learns, but also as the place where AI works. The user’s input and prompt go through the computers in data centres, where they get processed, and the AI hosted in those computers create an accurate output rapidly.

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## 3 Unsustainable practices of AI

### 3.1 Mining materials used for building servers

To build efficient servers and computers to enable and carry out AI operations, you need adequate equipment and components like various processors, semiconductors and other components, that are built from rare metals, minerals and other materials. This is not necessarily a problem tied only to AI industry, but to tech industry as a whole and its irresponsible consumers, who often update their technology mostly for prestige and to improve their image with new technology. But with the AI revolution, it is evident that AI also has a fair contribution towards the problem of unfair practices to the environment.

Copper is typically extracted from its ores through either pyrometallurgical or hydrometallurgical processes. In hydrometallurgy, chemicals like sulfuric acid are used to leach copper from oxide ores, while in pyrometallurgy, sulfide ores are smelted, releasing sulfur dioxide, a harmful air pollutant. Both methods pose a huge risk to environment, as the chemical waste can contaminate soil and water.

(Montgomery, 2025). Saad Rahim, chief economist of Trafigura branch in Switzerland, has put to attention at Financial Times Global Commodities Summit in Lausanne that the global demand for copper will add up to 1 million tons by 2030, with noticeable increase in this decade correlating with the rise of AI. (Desai, 2024).

Lithium is among the other metals used in producing equipment for AI, through lithium-ion batteries. Lithium-ion batteries do not directly enable AI operations, but the batteries provide the power to devices and infrastructure that are crucial for the development and deployment of generative AI. Cobalt, another popular semiconductor used in AI technology, enabling the efficiency of machines hosting and running AI. Unfortunately, cobalt mining is also not immune to devastating consequences as we can see on the example of Democratic Republic of Congo.

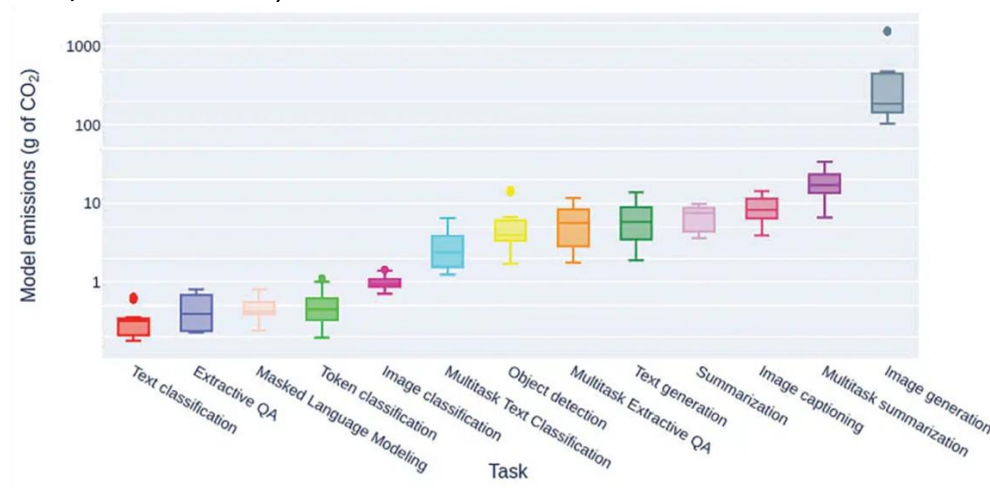
With the increase of popularity in AI industry, it is evident that the extraction of copper and lithium will only be increasing, and that the tech industry must resort to greener methods and recycled materials.

### 3.2 Power and CO<sub>2</sub> emissions

Asking a simple question on GPT models, instead of using a search engine like Google (with the AI assistant turned off), consumes 10 times more electricity. According to Sam Altman, CEO of OpenAI, typing a simple “thank you” or “please” in ChatGPT models is costing them millions of dollars. While it is just a few extra letters, that is also more work for the model to process. The more AI has to process and generate, the more power it will draw, thus creating a significant carbon footprint. In order to provide outputs, AI needs power to process and produce the requested prompt. The more complex the request is, more energy will be used, such as generating images which requires energy the most compared to other request such as simple text classification.

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Figure 1: Estimated CO<sub>2</sub> emissions from AI across different tasks (extracted from <https://www.polytechnique-insights.com/en/columns/energy/generative-ai-energy-consumption-soars/#note-content-8>)



Training AI models requires more power than it requires to deploy them to users. To successfully train a GPT model, it takes about 1300 megawatt hours (MWh) of electricity, with emitting 502 metric tons of CO<sub>2</sub>, which is equivalent to the annual consumption of 130 average households in USA (Vincent, 2024).

Most data centres run on energy from fossil fuels. It is estimated that the data centres account for up

to 3,7 percent of global CO<sub>2</sub> emissions, surpassing even the aviation industry. To avoid overheating and malfunctions, a good portion of electricity used in AI operations goes towards cooling systems. Data centres and servers have used air conditioning before, however liquid cooling is more energy efficient, using from 5 to 15 percent less energy and in recent years it is utilised more. (Cho, 2023) In 2021, data centres have accounted for 0,9-1,3 percent of global electricity demand, and one study estimated that number might reach 1,86 percent by 2030.

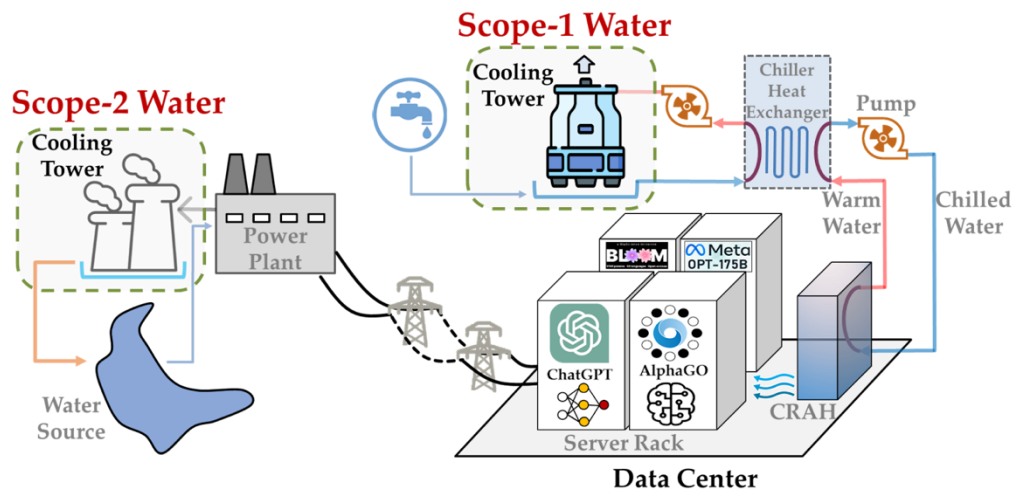
The solution in harm reduction could be the installation of solar panels and wind turbines of which data centres can benefit. While we as users cannot directly create company and local policies and what energy should be used, we can bring awareness to the problem and use less generative AI solutions, especially avoiding unnecessary questions that Google can answer.

### **3.3 Water**

Water often symbolizes life. Even if it's tasteless, odourless and provides no nutritional value, it is still vital for all living beings. Even if Earth is consisted of 71% of water, only 3% of water is fresh, with low concentrations of salt and other dissolved solids, making it drinkable. But only one third of fresh water is available, as two thirds are tucked away in frozen glaciers, icecaps or deep underground. Water shortage causes catastrophic events, environmental disruptions and can be detrimental to wildlife. The aquatic life is at risk of disappearing, disrupting their predators as well who feed of them.

The water also has a role in supporting AI operations through cooling its' physical infrastructure. As mentioned earlier, the core of the AI is found in data centres and servers used which consume a significant amount of power, which can cause overheating. To ensure that machines continue operating smoothly, they require temperature management via cooling systems, to avoid potential equipment failure. As mentioned earlier, throughout the history, air conditioning was used to cool down the servers and technology, however the density of the servers increased so air wasn't sufficient to continue cooling it down. Nowadays, most common way to cool down the data centres and servers is by using water, as the water cooling is generally more efficient than air cooling due to water's higher thermal conductivity and heat capacity, while consuming less power. This means water can absorb and dissipate heat more effectively than air, allowing for more compact and energy-efficient cooling systems. (Mytton, 2021)

Figure 2: An example of a data centre's operational water usage (extracted from <https://oecd.ai/en/work/how-much-water-does-ai-consume>)



It is estimated that ChatGPT consumes about 0,5 litres of water for every 5 to 50 prompts on average, without taking into account the complexity of the prompt. On the other hand, Google's data centres have consumed over 21 billion litres in 2022., which is a 20 percent increase from 2021, and Microsoft had the increase of 34% in the same period. (Gupta, 2024.) With increasing trends, it is clear that those figures will only rise even more, along with rising temperatures caused by CO2 emissions, which will affect water reserves and the demand for it. The experts estimate that AI will demand between 4,2 to 6,6 billion cubic meters of water by 2027, which is the equivalent of four to six times more than the country of Denmark annual withdrawal. (Ren, 2023).

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One of the less harmful practices is the use of desalinated and filtered ocean water in cooling, which some data centres are already practicing.

## 4 Conclusion

While the development of generative AI has been a major technological breakthrough that has eased many processes and the way we do our tasks, it has also been a contributing factor to the devastation of the environment. This research highlighted the environmental footprint caused by AI mostly with water pollution, CO2 emissions, endangering wildlife and withdrawing freshwater supplies either directly or indirectly. Data centres will continue to grow and have even more data in the future to work with, and they must resort to greener methods, like using renewable energy, recycled material and desalinated and filtered water. But unfortunately, real-life examples show us that main obstacle to businesses adapting to greener methods is orientation towards profit and increasing costs is not an option for many, not even for the sake of the planet, but some instead embrace AI and replace human workers in favour of it, mostly for the cost. The trends show us that further development of generative AI will not be ending at this point and models will only improve, causing even more environmental degradations.

However, we as regular citizens can avoid unnecessary use of AI tools, like treating it as a toy we can use when we are bored or resorting to AI to ask the simplest questions and participating in social media trends to see how we would look like as cartoon characters or how would our pets look like as humans.



Instead, we can use our imaginations and generate the image with our mind, pencils and hands. Avoiding practices that we know are harmful to our only home is not enough towards making a sustainable future and society, we need to raise awareness both among the civilians and also pressuring the tech companies, various organisations and the representatives of governments by writing them letters about the concerns to potentially propose further regulations within the industry. Technological advancement is inevitable, but it should not come at the cost of the nature.

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